Page No.
Date

- Schwarzschild metric ($G = c = 1$) in the equatorial plane ($\theta = \pi/2$)

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\phi^2$$

Let proper time be τ, overdots denote $d/d\tau$

The Lagrangian for geodesic motion:

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$$

a) Conserved quantities E and L:

Lagrangian:

$$\mathcal{L} = \frac{1}{2}\left[-\left(1 - \frac{2M}{r}\right)\dot{t}^2 + \left(1 - \frac{2M}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\phi}^2\right]$$

Cyclic Co-ordinates

- t doesn't appear explicitly $\longrightarrow$ Conserved energy
- ϕ doesn't appear explicitly $\longrightarrow$ Conserved Angular Momentum

Canonical momenta

$$p_t = \frac{\partial \mathcal{L}}{\partial \dot{t}} = -\left(1 - \frac{2M}{r}\right)\dot{t}$$

Define $E = -p_t = \left(1 - \frac{2M}{r}\right)\dot{t}$

Similarly,

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = r^2\dot{\phi} \Rightarrow L = p_\phi = r^2\dot{\phi}$$

Why $\theta = \pi/2$ is general:

By Spherical Symmetry, any geodesic can be rotated into the equatorial plane without changing the physics.

Page No
Date

(b) Radial Equation and Effective potential:

Normalization of 4-Velocity –

For timelike geodesics: $g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} = -1$

Substitute metric:

$$-\left(1-\frac{2M}{r}\right)\dot{t}^2 + \left(1-\frac{2M}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\phi}^2 = -1$$

Insert Conserved quantities:

$$\dot{t} = \frac{E}{1-2M/r}, \quad \dot{\phi} = \frac{L}{r^2}$$

After Simplification:

$$\dot{r}^2 + V^2(r) = E^2$$

Where

$$V^2(r) = \left(1-\frac{2M}{r}\right)\left(1+\frac{L^2}{r^2}\right)$$

This is the effective potential

(c) Radial free fall: $L = 0$

For this Experiment, Achilles falls Straight in:

$$V^2(r) = 1-\frac{2M}{r}$$

Initial Condition:

Starting from rest at $r = R$

$$\dot{r} = 0 \Rightarrow E^2 = 1-\frac{2M}{R}$$

Thus

$$\dot{r}^2 = 2M\left(\frac{1}{r} - \frac{1}{R}\right)$$

Cycloidal Solution:

Introduce parameter χ:

$$r(\chi) = \frac{R}{2}(1 + \cos\chi)$$

$$r(\chi) = \frac{R^{3/2}}{2\sqrt{2M}}(\chi + \sin\chi)$$

- $\chi = 0$: Start at $r = R$
- $\chi = \pi$: reach $r = 0$

Total proper time to reach the singularity

Set $\chi = \pi$:

$$\boxed{\tau_{fall} = \frac{\pi R^{3/2}}{2\sqrt{2M}}}$$

Finite proper time. Achilles reaches the Center.

(d) Co-ordinate time and the tortoise Co-ordinate

From energy Conservation:

$$\frac{dr}{dt} = \frac{\dot{r}}{\dot{t}} = \frac{(1 - 2M/r)\dot{r}}{E}$$

Define the tortoise Co-ordinate:

$$r_* = r + 2M \ln\left|\frac{r}{2M} - 1\right|$$

Pg No
Date

Differentiate:

$$\frac{dr_*}{dt} = \frac{\dot{r}}{\mathcal{E}}$$

Near the horizon $r \to 2M^+$:

From the radial equation:

$$\dot{r}^2 = \mathcal{E}^2 - V^2(r) \approx \mathcal{E}^2 \Rightarrow \dot{r} \approx -\mathcal{E}$$

Thus:

$$\frac{dr_*}{dt} \approx -1$$

Integrating:

$$r_*(t) \sim -t + \text{Const.}$$

Invert r_*:

$$r(t) \approx 2M + Ce^{-t/2M}$$

~~Important:~~

(e) No - not in Schwarzschild time

As $t \to \infty$

$$r(t) \to 2M$$

Achilles never reaches the event horizon in finite Co-ordinate time.

Page No.
Date

Resolution of the Achilles - Tortoise paradox:

- If observer = Achilles,
  It uses the proper time τ
  and crosses $r = 2M$ and reaches $r = 0$ in finite time.

- If observer = Distant observer
  It uses the Coordinate time 't'
  and Achilles asymptotically freezes at $r = 2M$

They both are correct Because:
- Proper time measures physical experience
- Schwarzschild time measures signals seen at infinity.
- The horizon is a Co-ordinate Singularity, not a physical one.

→ Achilles doesn't reach the tortoise - but only in his own time.

Just like Zeno's paradox:
- Infinite Subdivision in Coordinate time
- Finite completion in proper time.

General Relativity resolves Zeno Without Contradiction, by teaching us that time itself depends on the observer.